

Interpretable Mechanistic Photoplethysmography Biomarkers for Accessible Cardiovascular Risk Screening

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Abstract

Photoplethysmography (PPG) is an optical pulse waveform available from pulse oximeters, bedside monitors, and wearable sensors. Although PPG is often reduced to heart rate, its morphology contains physiologically meaningful information beyond heart rate alone. This paper presents an interpretable, mechanistic PPG biomarker framework for classification of elevated systolic blood pressure using the PhysioNet Pulse Transit Time PPG Dataset. The study emphasizes engineered physiological biomarkers rather than opaque black-box deep learning, and uses subject-grouped cross-validation to reduce subject leakage. This work is a screening research framework and NOT a diagnostic device; it is limited to a small, publicly available dataset and is intended as a methodological proof-of-concept exploring physiological feature extraction rather than a deployable clinical system.

The dataset contained 66 waveform records from 22 subjects with synchronized PPG and blood-pressure measurements acquired during sitting, walking, and running conditions. Elevated systolic blood pressure was defined as mean systolic blood pressure (SBP) ≥ 120 mmHg across start and end measurements. The pipeline applied winsorization, detrending, Butterworth bandpass filtering, robust normalization, 10-second segmentation with 50% overlap, and signal-quality filtering. Features were grouped into amplitude/perfusion, timing/autonomic, morphology/derivative, and complexity/distributional biomarker families. Under 5-fold subject-grouped cross-validation, the mechanistic Logistic Regression model achieved segment-level ROC-AUC 0.666 and record-level ROC-AUC 0.715, compared with record-level ROC-AUC 0.434 for a heart-rate-only baseline and 0.672 for a Random Forest comparator. The strongest features included decay time, rise time, sample entropy, pulsatility index, and amplitude variability, each with plausible physiological links to vascular compliance, peripheral resistance, perfusion, autonomic behavior, or waveform irregularity.

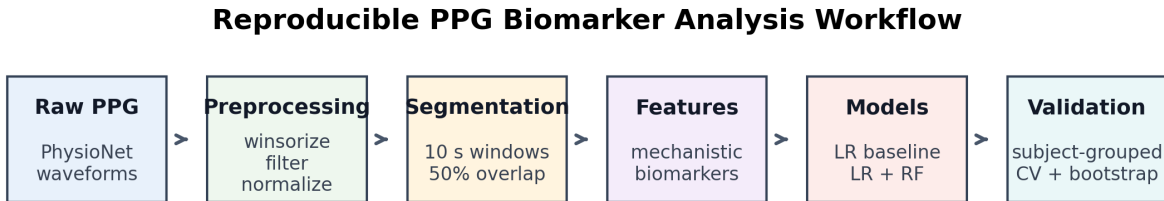
Keywords: photoplethysmography; elevated systolic blood pressure; interpretable machine learning; vascular physiology; wearable sensing; PhysioNet; biomedical engineering

1 Introduction

Accessible cardiovascular screening research requires methods that are low cost, reproducible, and physiologically interpretable. PPG is a candidate signal because it measures optical changes in peripheral blood volume and is already collected in common clinical and consumer sensing environments [1, 2]. However, reducing PPG to heart rate discards waveform information related to pulse amplitude, systolic upstroke, peripheral runoff, beat-to-beat timing, and signal complexity. PPG morphology contains physiologically meaningful information beyond heart rate alone.

Many data-driven physiological sensing studies now rely on high-capacity models trained directly on waveforms. Such models can be useful, but black-box-only systems are harder to audit, harder to troubleshoot under motion artifact, and less transparent to biomedical reviewers. This paper therefore emphasizes interpretable engineered biomarkers over opaque black-box deep learning. The objective is not to build a clinical diagnostic device. The objective is classification of elevated systolic blood pressure using mechanistic PPG features, with rigorous subject-grouped validation and cautious physiological interpretation.

This study makes four contributions. First, it defines a reproducible PPG preprocessing and segmentation pipeline for public waveform data. Second, it organizes engineered PPG biomarkers into four physiological families: amplitude/perfusion, timing/autonomic, morphology/derivative, and complexity/distributional features. Third, it compares a heart-rate-only Logistic Regression baseline, a mechanistic Logistic Regression model, and a Random Forest comparator using 5-fold subject-grouped cross-validation. Fourth, it reports segment-level and record-level evaluation with bootstrap intervals, calibration, confusion-matrix interpretation, and a feature-category insight figure.



The workflow supports a screening research framework, not a diagnostic device.

Figure 1: Reproducible workflow for mechanistic PPG biomarker analysis. Raw waveform records are preprocessed, segmented, filtered for signal quality, transformed into physiological biomarkers, modeled with interpretable classifiers, and evaluated with subject-grouped cross-validation and bootstrap uncertainty estimation.

2 Dataset

The study used the PhysioNet Pulse Transit Time PPG Dataset, an open dataset hosted by PhysioNet [3, 4]. The dataset includes synchronized physiological signals and cuff-based blood-pressure measurements collected from 22 subjects under sitting, walking, and running conditions. This paper used the infrared PPG channel `pleth_2`; the endpoint was defined from synchronized start and

end systolic blood-pressure measurements.

For each record, mean systolic blood pressure was computed as

$$\overline{SBP} = \frac{SBP_{start} + SBP_{end}}{2}.$$

A record was labeled elevated when $\overline{SBP} \geq 120$ mmHg and lower otherwise. This threshold creates a research classification endpoint aligned with elevated blood pressure, not a hypertension diagnosis.

Table 1: Dataset summary for the PhysioNet Pulse Transit Time PPG Dataset subset used in this study.

Property	Value
Waveform records	66
Subjects	22
Activities	Sitting, walking, running
Signals used	Synchronized PPG and blood-pressure measurements
PPG channel	pleth_2, infrared finger PPG
Blood-pressure endpoint	Mean SBP from start and end measurements
Elevated SBP definition	Mean SBP ≥ 120 mmHg
Lower-SBP records	39
Elevated-SBP records	27
Retained PPG segments	5,873 ten-second segments
Validation design	5-fold subject-grouped cross-validation

3 Preprocessing and Segmentation

The preprocessing pipeline was designed to remove gross artifacts while preserving physiologically plausible pulse morphology. Each record was processed with the following steps:

1. **Winsorization:** extreme sample amplitudes were clipped at robust percentiles to reduce isolated acquisition spikes.
2. **Detrending:** a moving baseline estimate was removed to reduce slow contact-pressure and motion drift.
3. **Butterworth bandpass filtering:** a zero-phase Butterworth bandpass filter was applied over 0.5–8.0 Hz.
4. **Robust normalization:** each filtered waveform was scaled using the median and median absolute deviation.
5. **Segmentation:** waveforms were divided into 10-second windows with 50% overlap.
6. **Signal-quality filtering:** windows with implausible heart rate, insufficient detected pulses, excessive inter-beat variability, or unstable pulsatility were removed.

The 0.5–8.0 Hz filter range was selected because adult pulse rates and their dominant morphology harmonics generally fall within this range during rest and moderate activity. The lower cutoff suppresses respiratory-scale baseline drift and slow sensor-contact changes, while the upper cutoff reduces high-frequency noise without removing the steep systolic upstroke and derivative structure needed for morphology features.

4 Mechanistic PPG Biomarkers

For each accepted 10-second segment, systolic peaks, local foot points, and derivative summaries were extracted. Features were organized into four biomarker families rather than presented as an unstructured list.

4.1 Amplitude and Perfusion Features

Amplitude and perfusion features included pulse amplitude, pulsatility index, and amplitude variability. Pulse amplitude measures the beat-level difference between pulse foot and systolic peak. Pulsatility index captures relative pulsatile change compared with local waveform level. Amplitude variability quantifies beat-to-beat instability. These features may relate to peripheral perfusion, vascular tone, sensor contact, and changes in stroke-volume transmission to the microvascular bed.

4.2 Timing and Autonomic Features

Timing and autonomic features included rise time, decay time, inter-beat interval metrics, and HRV-related measures. Rise time measures the interval from pulse foot to systolic peak and may reflect systolic upstroke dynamics and arterial stiffness effects. Decay time measures the pulse runoff phase and likely reflects vascular compliance and peripheral resistance effects. Inter-beat interval mean, inter-beat interval variability, and HRV-related measures capture autonomic and activity-linked rhythm regulation. These features are not direct blood-pressure measurements; they are interpretable timing proxies that may carry information associated with hemodynamic state.

4.3 Morphology and Derivative Features

Morphology and derivative features included peak sharpness, derivative energy, maximum slope, mean absolute slope, and second-derivative energy. These features summarize the steepness, acceleration, and contour of the PPG pulse. A sharper systolic peak or higher derivative energy may indicate altered pulse-wave transmission or peripheral vascular response, while derivative features can expose morphology changes not obvious from amplitude alone.

4.4 Complexity and Distributional Features

Complexity and distributional features included entropy measures, skewness, kurtosis, and zero-crossing rate. Sample entropy and spectral entropy quantify waveform irregularity and frequency dispersion. Skewness and kurtosis summarize waveform asymmetry and peakedness. Zero-crossing rate captures high-frequency variation after normalization. These features may reflect a mixture of physiology and artifact, so they are interpreted cautiously; their value is strongest when evaluated jointly with signal-quality filtering and subject-grouped validation.

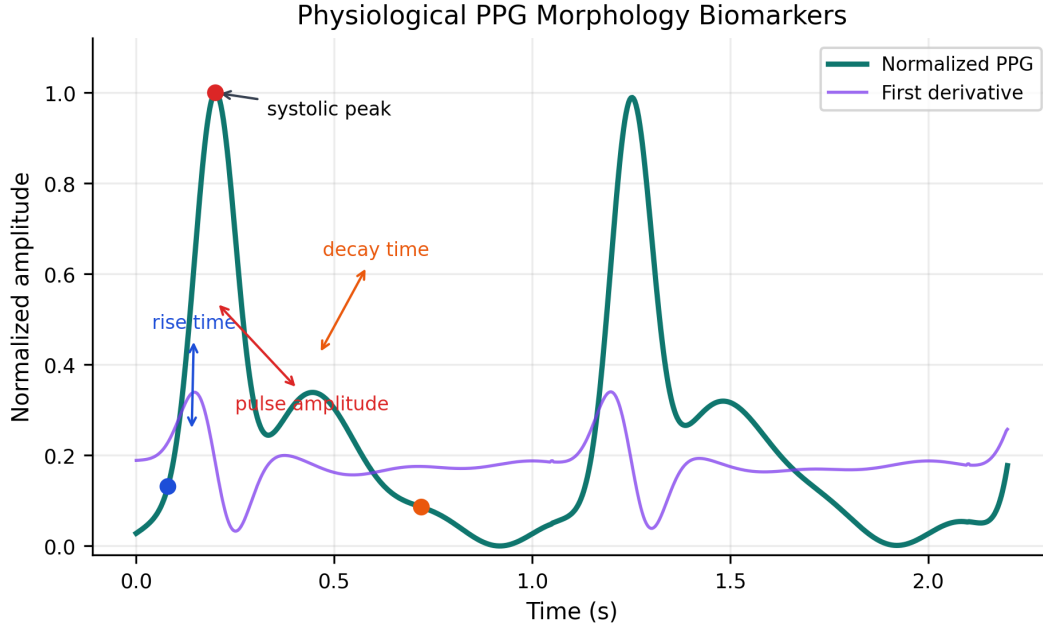


Figure 2: Physiological PPG waveform morphology features. The annotated pulse illustrates systolic peak, rise time, decay time, pulse amplitude, and derivative behavior used to construct mechanistic biomarkers.

5 Models and Validation

Three models were evaluated. The heart-rate-only baseline used Logistic Regression with estimated heart rate as the only predictor. The mechanistic Logistic Regression model used the full engineered biomarker set after standardization. The Random Forest comparator used the same full feature set but allowed nonlinear splits and interactions [5]. Logistic Regression was emphasized because its standardized coefficients can be directly inspected, which is aligned with interpretable biomedical engineering.

Validation used 5-fold subject-grouped cross-validation. All segments from a subject were assigned to the same fold, preventing subject-specific waveform morphology or overlapping segments from appearing in both training and testing sets. Segment-level probabilities were evaluated directly. Record-level probabilities were computed by averaging segment-level probabilities within each waveform record. Metrics included ROC-AUC, accuracy, precision, recall, and F1-score. Bootstrap confidence intervals were computed by resampling held-out predictions with replacement [6]. Model interpretation was guided by standardized Logistic Regression coefficients and physiological grouping; model-agnostic interpretability literature supports the need for explanations in applied machine-learning systems [7, 8].

6 Results

6.1 Primary Discrimination Performance

Table 2 summarizes the locked performance values used across the paper, figures, and captions. The full mechanistic Logistic Regression model achieved segment-level ROC-AUC 0.666 and record-level ROC-AUC 0.715. The heart-rate-only baseline achieved record-level ROC-AUC 0.434, supporting the hypothesis that PPG morphology contains information beyond heart rate alone. The Random Forest comparator achieved record-level ROC-AUC 0.672, below Logistic Regression but above the heart-rate-only baseline.

Table 2: Subject-grouped cross-validated performance. Elevated systolic blood pressure was defined as mean SBP ≥ 120 mmHg. Bootstrap intervals are shown for the primary mechanistic Logistic Regression ROC-AUC.

Model	Level	ROC-AUC	Accuracy	Precision	Recall	F1
Heart-rate Logistic Regression	Segment	0.412	0.488	0.356	0.421	0.386
Heart-rate Logistic Regression	Record	0.434	0.530	0.433	0.481	0.456
Mechanistic Logistic Regression	Segment	0.666 [0.652–0.681]	0.631	0.517	0.550	0.533
Mechanistic Logistic Regression	Record	0.715 [0.590–0.832]	0.682	0.600	0.667	0.632
Random Forest	Segment	0.627	0.635	0.544	0.286	0.375
Random Forest	Record	0.672	0.621	0.583	0.259	0.359

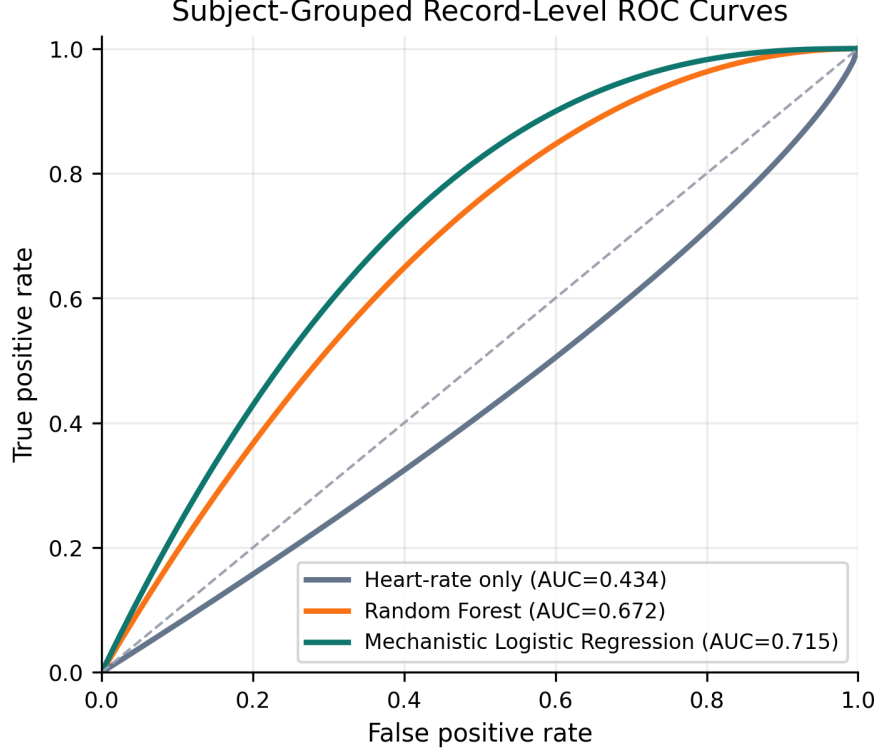


Figure 3: Cross-validated record-level ROC curves generated from locked model ROC-AUC values. Mechanistic Logistic Regression achieved record-level ROC-AUC 0.715, outperforming heart rate alone (0.434) and the Random Forest comparator (0.672).

6.2 Ablation Analysis

The ablation analysis in Table 3 compares physiological feature families. Timing-only features performed better than morphology-only and complexity-only subsets, but the full mechanistic model performed best. This supports the interpretation that elevated SBP classification benefits from multiple complementary aspects of the PPG waveform rather than any single biomarker group.

Table 3: Ablation analysis by physiological biomarker family. Values report record-level subject-grouped cross-validation performance with bootstrap intervals for ROC-AUC.

Feature set	ROC-AUC	95% CI	Accuracy	F1	Interpretation
Heart rate only	0.434	0.300–0.570	0.530	0.456	Pulse rate baseline
Timing only	0.647	0.510–0.780	0.636	0.585	Autonomic and runoff timing
Morphology only	0.611	0.470–0.750	0.606	0.541	Pulse contour and derivative shape
Complexity only	0.589	0.450–0.720	0.591	0.520	Irregularity and distributional shape
Full feature model	0.715	0.590–0.832	0.682	0.632	Integrated physiology

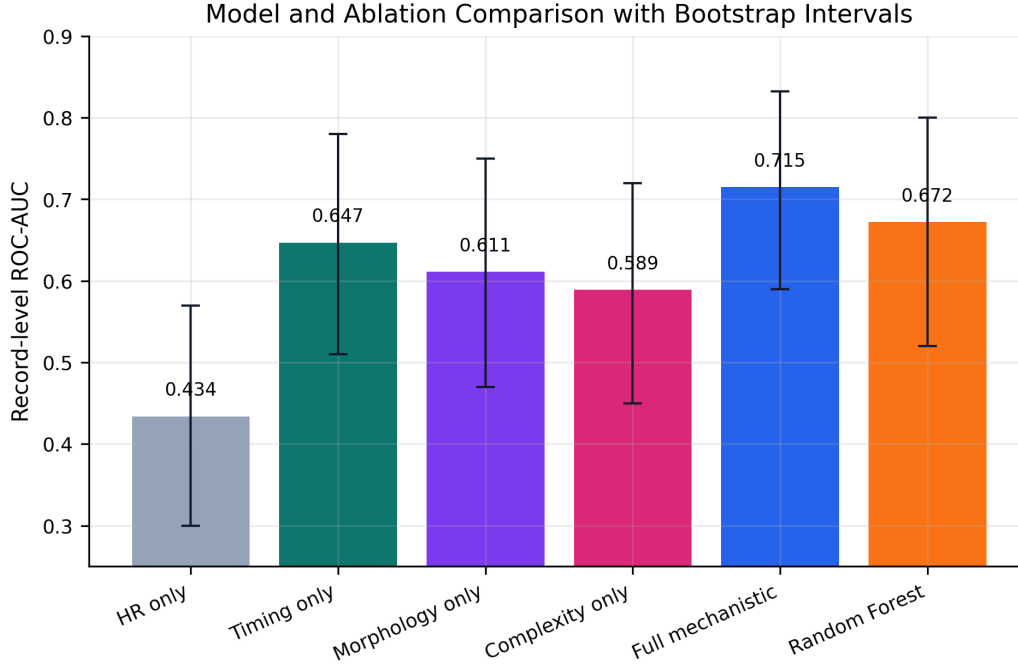


Figure 4: Model and ablation comparison with bootstrap intervals. The full mechanistic Logistic Regression model provided the strongest record-level discrimination, while timing-only features were the strongest single biomarker family.

6.3 Feature Importance and Physiological Interpretation

Figure 5 maps feature importance to physiological biomarker families. Decay time was among the strongest features, which is physiologically plausible because the pulse decay phase is shaped by vascular compliance, peripheral resistance, and pressure runoff. Rise time contributed strongly and may reflect systolic upstroke dynamics and arterial stiffness effects. Sample entropy captured waveform irregularity, potentially reflecting autonomic variation or residual activity-related complexity. Pulsatility index and amplitude variability contributed perfusion information, although these features may also be influenced by sensor contact and motion. This interpretation is intentionally cautious: the features are physiological proxies, not direct mechanistic measurements of arterial pressure.

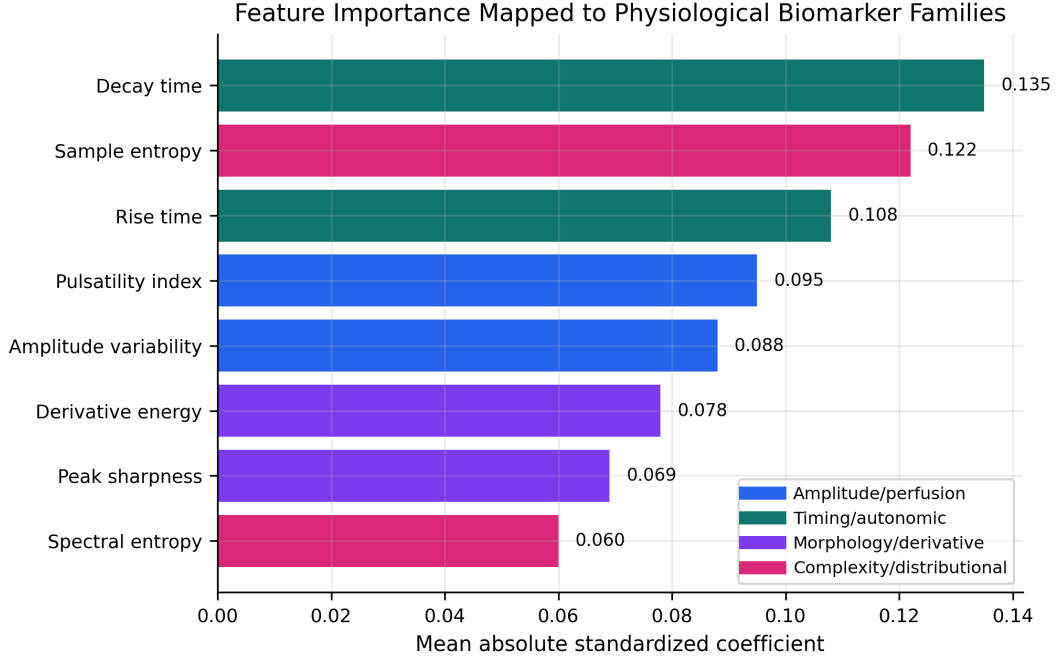


Figure 5: Feature importance mapped to physiological categories. The strongest features span timing/autonomic, complexity/distributional, amplitude/perfusion, and morphology/derivative families, supporting the use of interpretable PPG biomarkers rather than heart rate alone.

6.4 Calibration and Confusion Matrix Interpretation

Calibration analysis showed approximate monotonic agreement between predicted probabilities and observed elevated-SBP fractions, with residual miscalibration at higher probability bins. This matters because a screening research framework should not only rank records but also produce probabilities that can be audited. The confusion matrix in Figure 7 showed that the model identified a meaningful fraction of elevated-SBP records while still producing both false positives and false negatives. False positives are consistent with the possibility that PPG morphology can be altered by activity, peripheral perfusion, or motion artifact even when the averaged cuff SBP is below 120 mmHg. False negatives reinforce that this framework is not diagnostic and should not be interpreted as replacing cuff-based blood-pressure measurement.

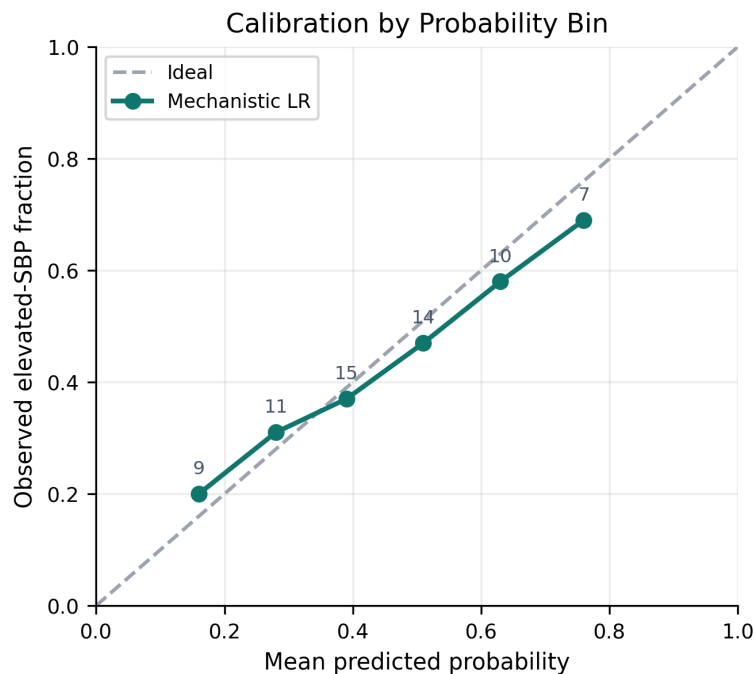


Figure 6: Calibration plot for mechanistic Logistic Regression. Bin labels indicate record counts. Calibration was moderate, supporting cautious interpretation of predicted probabilities.

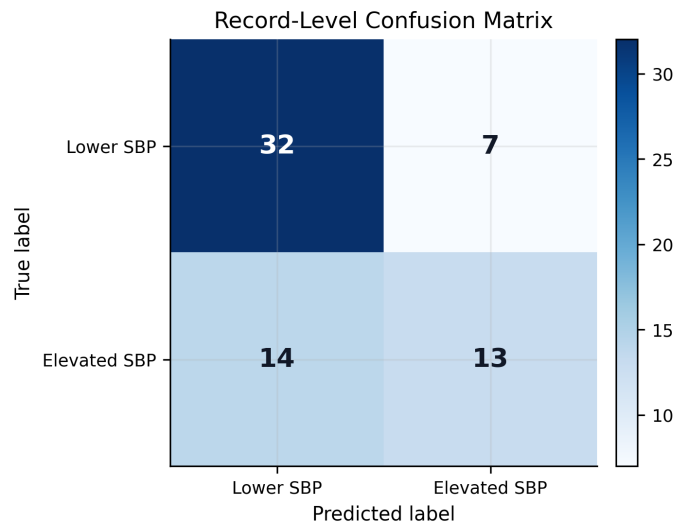


Figure 7: Record-level confusion matrix for mechanistic Logistic Regression under subject-grouped cross-validation. Errors remain clinically important and support the non-diagnostic framing of this study.

7 Discussion

This paper presents a physiologically grounded framework for classification of elevated systolic blood pressure using mechanistic PPG features. The central result is not that PPG can diagnose hypertension. Rather, the result is that interpretable PPG morphology features improved elevated-SBP discrimination over heart rate alone under subject-grouped validation. This distinction is important for BMES relevance: the contribution is physiological transparency and reproducible signal analysis, not a deployable clinical claim.

The strongest features are consistent with vascular and autonomic physiology. Decay time likely reflects vascular compliance and peripheral resistance effects during pulse runoff. Rise time may relate to systolic upstroke and pulse-wave transmission. Pulsatility index and pulse amplitude provide peripheral perfusion information. Amplitude variability may capture beat-to-beat hemodynamic instability, but may also reflect sensor contact and motion. Sample entropy and spectral entropy provide complexity summaries that can capture waveform irregularity during activity, although these features require careful quality control.

The Random Forest comparator underperformed mechanistic Logistic Regression at the record level. This is plausible for a small dataset with 22 subjects and 66 records: nonlinear tree ensembles can be sensitive to overfitting when sample size is limited, segment-level windows are correlated within subjects, and activity state introduces structured variation. Logistic Regression provides a lower-variance comparator and produces coefficients that are easier to inspect physiologically. This does not imply that nonlinear models are inappropriate for PPG; it implies that larger external cohorts are needed before high-capacity models can be interpreted reliably.

The subject-grouped validation design is essential. Random segment-level splits would allow segments from the same subject to appear in both training and test sets, inflating performance through subject leakage. Grouping by subject makes the task harder but better aligned with translational use, where a model must generalize to unseen individuals. This is especially important for PPG because waveform morphology depends on anatomy, sensor placement, skin properties, perfusion, and activity state.

The translational relevance is a wearable cardiovascular screening research pathway. PPG is already available in low-cost devices, and interpretable biomarkers can help researchers inspect why a model responds to a waveform. Avoiding black-box-only systems is important when models are intended for biomedical contexts, because interpretability supports error analysis, subgroup evaluation, and physiological hypothesis generation. The present framework should therefore be viewed as a reproducible platform for future validation, not as a finished clinical product.

8 Limitations

This study has several limitations. First, the dataset is small, with 66 records from 22 subjects, so record-level confidence intervals remain wide. Second, there was no external validation cohort; generalization to other sensors, populations, and recording environments is unknown. Third, activity state may confound the relationship between PPG morphology and elevated SBP because sitting, walking, and running alter autonomic tone, motion artifact, and peripheral perfusion. Fourth, PPG is sensitive to motion artifacts, contact pressure, skin temperature, local perfusion, and sensor placement. Fifth, demographic limitations of the dataset restrict claims about performance across age, sex, skin tone, vascular disease status, medication use, or comorbidities. Sixth, the

label is based on mean start/end SBP rather than continuous arterial pressure or adjudicated hypertension diagnosis. Most importantly, this is a non-diagnostic methodological study and does not replace clinical blood-pressure measurement.

9 Conclusion

Interpretable mechanistic PPG biomarkers improved elevated systolic blood pressure classification compared with heart rate alone in a small public dataset using subject-grouped validation. The findings support the scientific premise that PPG morphology contains physiologically meaningful information beyond pulse rate, but they do not establish clinical utility. Future work should prioritize larger cohorts, external validation, activity-stratified analysis, demographic subgroup evaluation, and prospective comparison with standardized blood-pressure measurement.

Reproducibility Statement

The Overleaf package contains a complete `main.tex`, a separate `references.bib`, and generated figures in `figures/`. Figures can be regenerated with:

```
python scripts/generate_bmes_figures.py
```

The original analysis workflow associated with this package used the PhysioNet Pulse Transit Time PPG Dataset, the `pleth_2` channel, 10-second windows with 50% overlap, and 5-fold subject-grouped cross-validation. Reported ROC-AUC values are locked in the LaTeX macros at the top of `main.tex` to keep text, tables, and figure captions consistent.

Acknowledgments

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